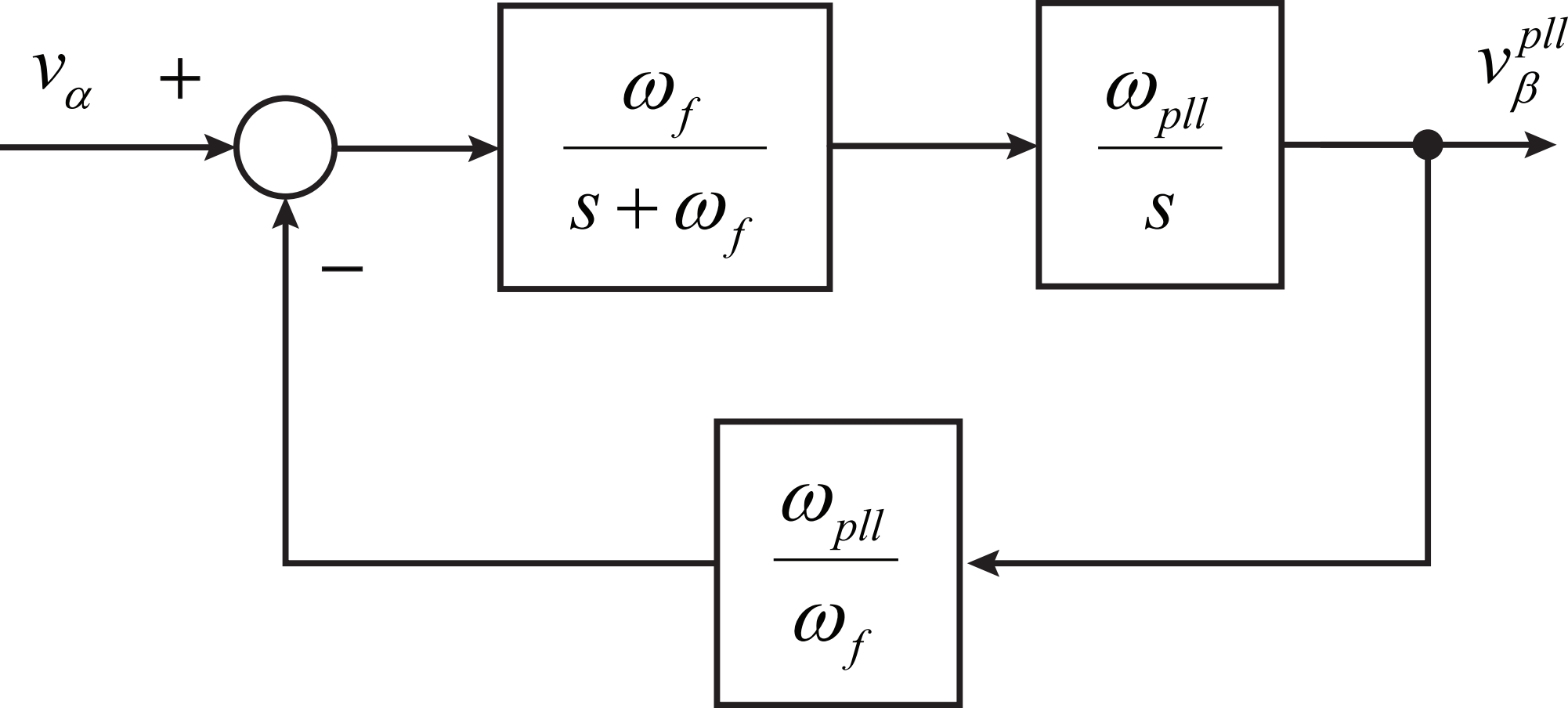
v_α
+
−
$\frac{\omega_f}{s+\omega_f}$
$\frac{\omega_{pll}}{s}$
v_β^{pll}
$\frac{\omega_{pll}}{\omega_f}$